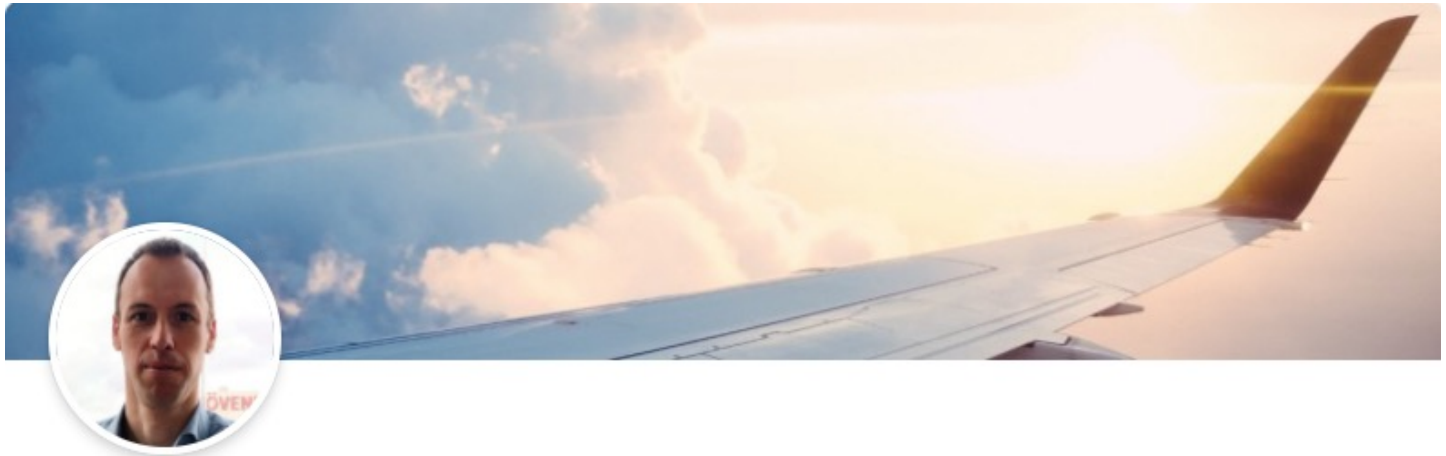
UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Data Science in Online Marketing

Course Introduction

Maarten Soomer

Let me introduce myself..



Maarten Soomer

Lead Data Scientist @ Travix & Assistant Professor @ University of Amsterdam

Assistant Professor
University of Amsterdam
Mar 2015 - Present

Lead Data Scientist
Travix International
Feb 2020 - Present

CheapTickets.nl BudgetAir FlugLaden Vayama

University of Amsterdam
Amsterdam Business School

Data Science in Online Marketing

FINANCIAL TIMES

Tech turns advertising's 'Mad Men' into maths men

Robert Cookson

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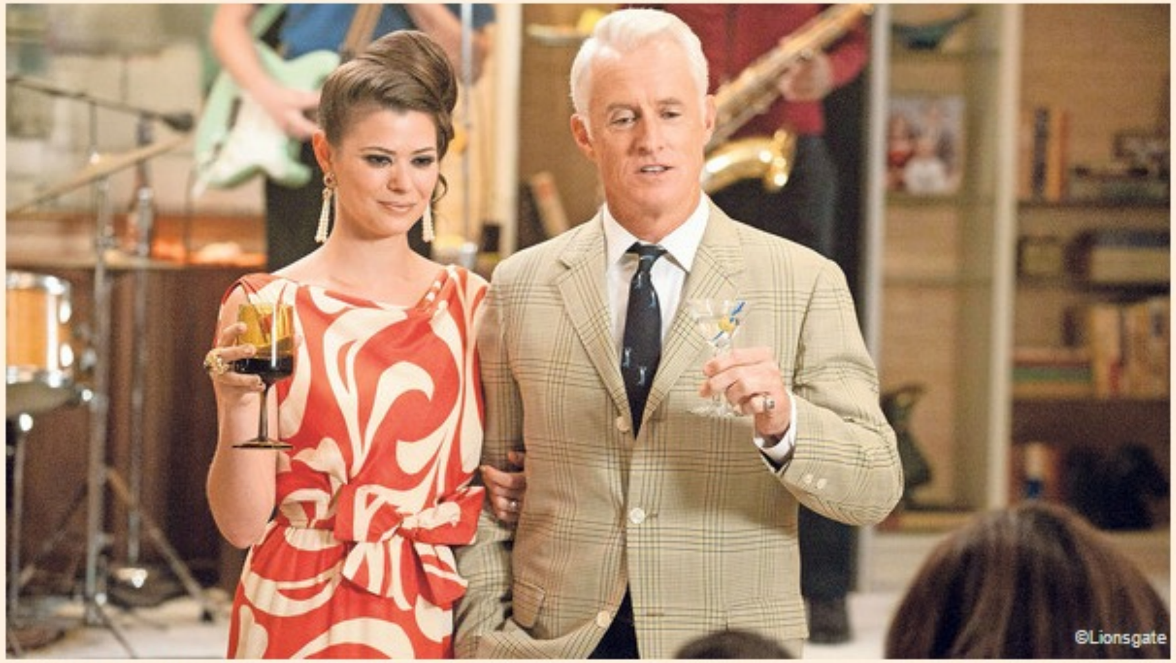


John Slattery as Roger Sterling in Mad Men

An online advertising campaign last year by Kronenbourg, a French brewer, adopted an innovative tactic: it used real-time temperature data to target only those people it calculated would be most susceptible to the temptation of a refreshing beer.

To execute the campaign, Kronenbourg worked with Infectious Media, a UK-based agency that specialises in the fast-growing area of "programmatic" advertising, or using automated computer systems to deliver ads across the internet.

Image taken from: Financial Times, ft.com

Driving force: data

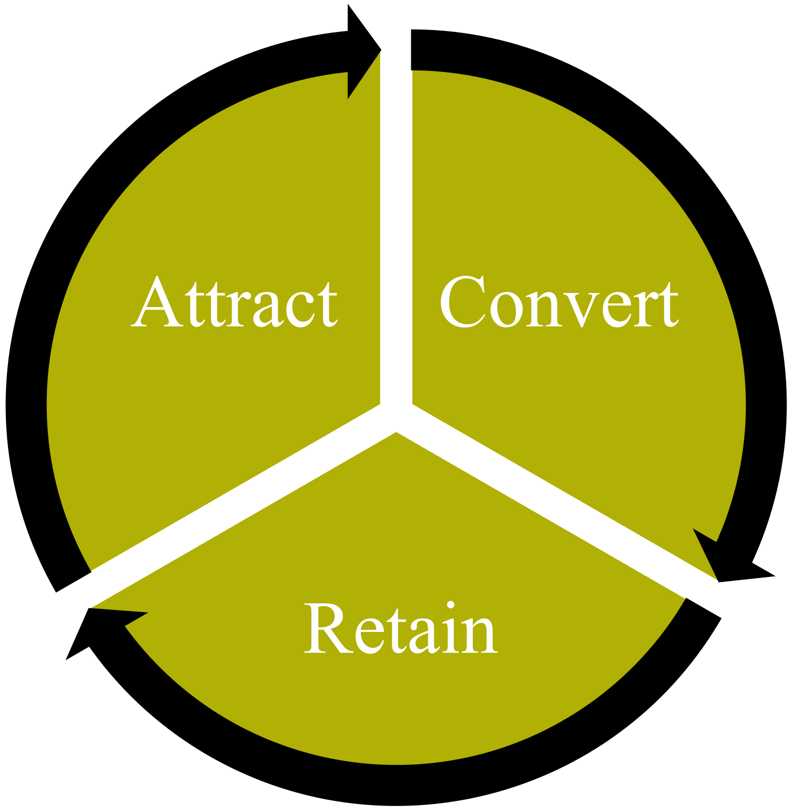
Who knows what to do with all that data?

Image taken from: Haravrd Business Review, hbr.org

What?

The right product
for the right customer
at the right time
at the right place
for the right price
with the right message (ad)
effectively and efficiently

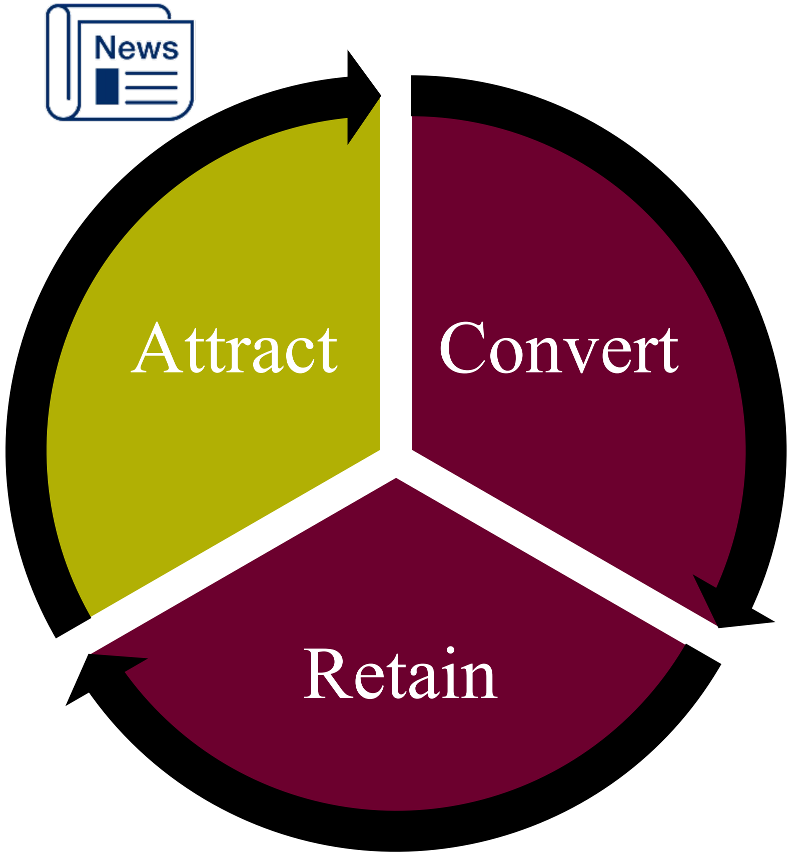
E-commerce sales cycle



E-commerce sales cycle

Marketing & advertising to attract (new) customers:

- Direct Cost involved: Marketing Investment
- Large **online advertising** budgets: How to spend effectively?

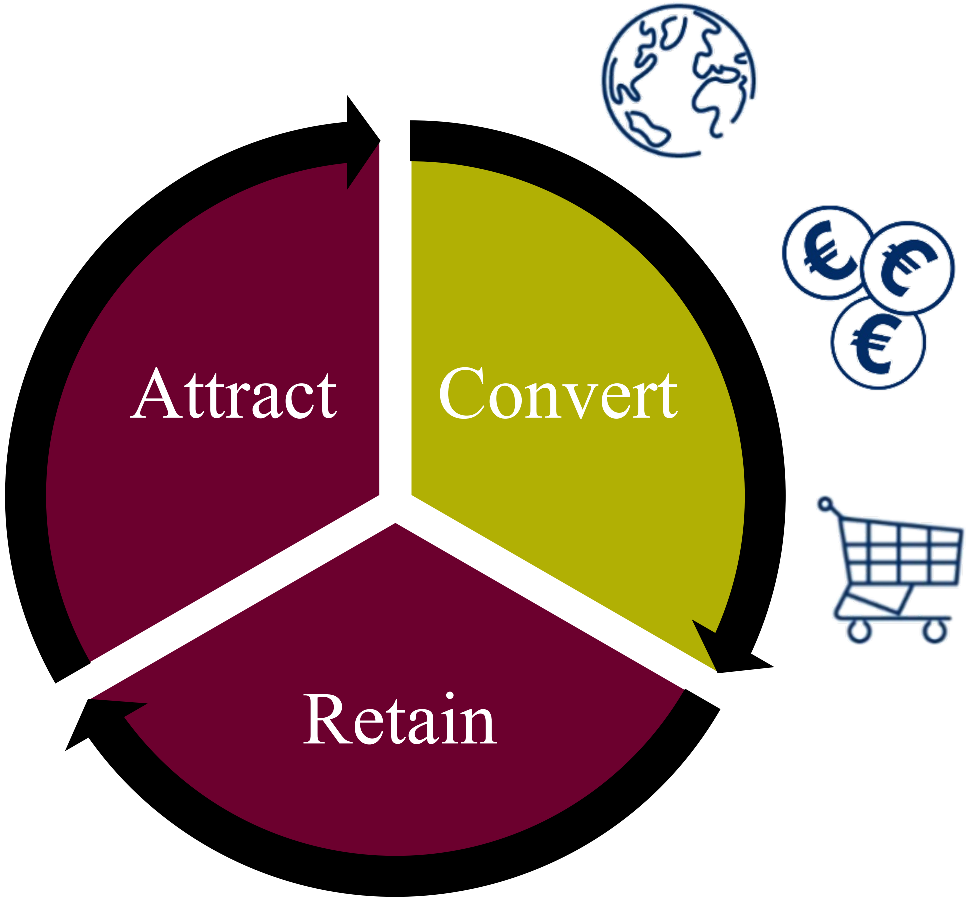


E-commerce sales cycle

Convert visitors of e-commerce site:

- Relevant Product Offering
- **Relevant, personalized & optimized website**
- Competitive Price

Make sure marketing investment is not wasted by a bad user experience



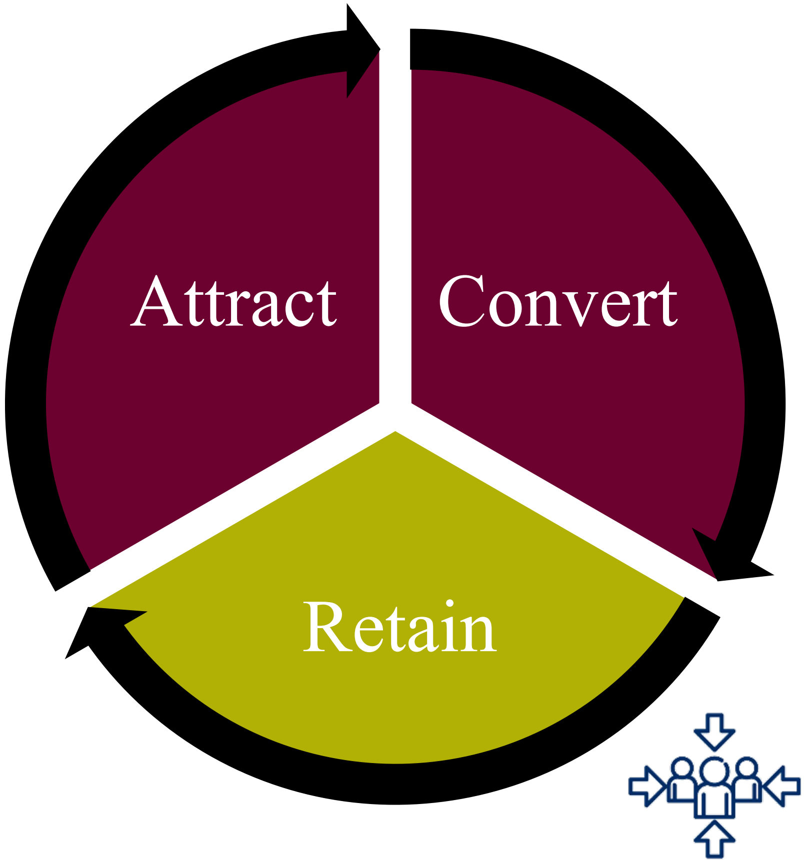
E-commerce sales cycle

Retain existing customers:

- Excellent Service
- **Relevant, personalized recommendations** and communication

Increase customer lifetime value and minimize advertising investment need to re-activate customers.

Use your knowledge of the customer.



Data Science in Online Marketing Course

Get a deep dive on concepts and algorithms used in online marketing:

Online Advertising

- CTR prediction
- Media Mix Modeling & Conversion Attribution

Recommenders

- Content Based
- Collaborative Filtering

Data Science in Online Marketing Course

Hands-on **computer assignments**, putting it all together:

- Working with real life big(ger) data sets
- Using Python and Jupyter Notebooks (Google Colab)
- Building on what you have learnt in Coding Lab, Machine Learning courses

Data → Exploratory Data Analysis → Predictive Model

Course Outline

Date	Topic	Assignment
3 april	**Recommenders: Content Based**	
10 april	Recommenders: Collaborative Filtering	Yes
17 april	Recommenders: Guest Lecture Bol.com Online advertising: Introduction	
24 april	Online advertising: Reinforcement Learning	
1 may	Online advertising: CTR prediction	Yes
15 may	Online advertising: Media Mix Modeling & Conversion Attribution	
22 may	Revision	
29 may	Digital exam on location	

Lecture Setup

- 3-hour lecture each week

- 18:30 - 19:30: presentation of lecture notes
- 19:45 - 20:45: presentation of lecture notes
- 21:00 - 22:00: practice exam style questions/discuss assignments

- For each lecture, some exam style questions are made available on Canvas
- You can practice them during the 3rd hour of the lecture
- Answers are discussed the next week during the 3rd hour of the lecture

Group Assignments & Grading

- 2-3 students per group
- 2 computer assignments

- The final grade is based on the following elements:
  - ❑ Written exam (60%)
  - ❑ Group Assignment 1 (20%)
  - ❑ Group Assignment 2 (20%)
- In order to pass the course, the written exam grade must be 5.5 or higher and the weighted course average must be 5.5 or higher. In case of a resit, only the written exam can be retaken. The results obtained for group case work remain valid for the resit.

Computer Assignments Guidelines

- Analysis and modeling on a dataset
- Submit a (concise) report and notebook (e.g., Google Colab link)

- Report: Do not only write what you did, but also
  - ❑ Why you did it: Motivate your assumptions and choices
  - ❑ What are the results: include some charts/tables of the most important results in the report
  - ❑ Interpretation of the results: How do you interpret the results? What conclusions do you have and what's the impact for next steps?
- Code: Do not copy (major) code snippets/texts from online sources
  - ❑ **Always cite all the resources you reference**
  - ❑ In case of exceptions/doubt: reference and motivate

Exam

- Digital (TestVision)
- Physical on a university location
- University supplies laptops
- You are allowed to use:
  - ❑ Calculator (simple, no internet connection)
  - ❑ Formula sheet (supplied by university)
  - ❑ Scratch paper (supplied by university)
- You are NOT allowed to use:
  - ❑ Connected devices: phones etc.
  - ❑ Other applications, browser tabs on the laptop